GATE

Computer Science & Information Technology

Computer Networks

Application Layer Protocols

DPP: 1

Q1 In the following pairs of OSI protocol layers and their functionality, which one is the CORRECT pair / s?
1. Network layer and Routing
2. Data Link Layer and Bit synchronisation
3. Transport layer and End-to-end process communication
4. Medium Access Control sublayer and Channel sharing
(A) 1 & 2 only
(B) 2 & 3 only
(C) 1, 2 & 3
(D) 1, 3 & 4 only

Q2 In the following pairs of OSI protocol layer and their functionality, which one is the CORRECT pair / s?
1. Segmentation and reassembly are done at hosts only, not at each intermediate router.
2. The Application Layer allows the user or application to access the network.
3. The Session Layer establishes, maintains, and synchronises the interaction between the communicating systems.
4. The Presentation Layer allows the process to add checkpoints or synchronisation points in the stream of data.
(A) 1 & 2 only (B) 2 & 3 only
(C) 1, 2 & 3 (D) 3 & 4 only

Q3 Which of the following is/are CORRECT?
1. Physical addresses in the packet are modified from HOP to HOP.
2. Network Layer ensures packets belonging to the same message arrive intact and in order.
3. The network layer uses physical addresses for addressing.
4. The Transport Layer treats every packet individually.
(A) 1 only (B) 2 & 3 only
(C) 1, 2 & 3 (D) 3 & 4 only

Q4 Choose the correct statement/s:
1. The unit of communication at the physical layer is a bit.
2. The unit of communication at the network layer is a datagram.
3. The unit of communication at the transport layer is a segment.
4. The unit of communication at the transport layer is a user datagram or a packet.
(A) 1 & 2 only
(B) 2 & 3 only
(C) 1, 2 & 3
(D) 1, 2, 3 & 4

Q5 Choose the incorrect statement/s:
1. The computer used for forwarding IP packets is called an IP router
2. The Session Layer decides whether the way of communication between them is Half Duplex or Full Duplex.
3. The physical addresses will change from hop to hop, but the logical and port addresses usually remain the same
4. The two most typical network applications that

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use UDP are File Transfer Protocol (FTP) and the TELNET

(A) 1 & 2 only (B) 2 & 3 only

(C) 1 & 3 only (D) 4 only

Q6 Match the protocols from Group I to one or more in Group II:

	Group		Group II (port, TCP/UDP)
1.	TELNET	A.	23 TCP
2.	SMTP	B.	25 TCP
3.	DHCP SERVER	C.	67 UDP
4.	FTP - DATA	D.	20 TCP
5.	FTP - CONTROL	E.	21 TCP
		F.	68 UDP
6.	DHCP CLIENT	G.	67 TCP
		H.	68 TCP

(A) 1- A,2-B,3-C,4-D,5-E,6-F

(B) 1- A,2-B,3-C,4-D,5-E,6-F

(C) 1- A,2-B,3-C,4-D,5-E,6-F

(D) 1- A,2-B,3-C,4-D,5-E,6-F

Q7 Which layer of the ISO-OSI layer is responsible for the synchronisation of the stream of data?

(A) Session Layer

(B) Presentation Layer

(C) Network Layer

(D) Data Link Layer

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Answer Key

Q1	(D)
Q2	(C)
Q3	(A)
Q4	(D)
Q5	(D)
Q6	(A)
Q7	(A)

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Hints & Solutions

Q1 Text Solution:

One of the main functions of the Network Layer is Routing. So it is CORRECT.

Bit Synchronization is always handled by the Physical Layer of the OSI model but not the Dat Link Layer. So, it is INCORRECT.

The transport layer handles end-to-end process communication.

So it is CORRECT.

MAC sublayer has 3 types of protocols (Random, Controlled and Channelized Access)

Q2 Text Solution:

True, Segmentation and reassembly are functionalities of the Transport Layer. The transport layer is absent at routers. So, segmentation and reassembly are done only at the source or destination.

True, the Application Layer provides applications to users to access network resources.

True, the Session Layer establishes, maintains, and synchronises the interaction between the communicating systems. Session Layer allows the process of adding checkpoints or synchronisation points to the stream of data.

False, Presentation Layer handles the syntax and semantics of information sent.

Q3 Text Solution:

True, the MAC addresses of the next HOP are updated on every intermediate router.

False, at Network Layer all the packets belonging to the same message are treated as different packets.

False, The Network layer uses logical addresses for addressing.

False, Transport Layer ensures that the whole message arrives intact and in order.

Q4 Text Solution:

Physical Layer - Bit

Data Link Layer - Frame

Network Layer - Datagram

Transport Layer - Segment or Packet and User Datagram, based on the protocols TCP and UDP, respectively.

Application Layer - Message

Q5 Text Solution:

Option 1, 2, and 3 are all True,

The only option is 4 false.

Reliability is key for FTP and TELNET applications, So these applications use TCP.

Q6 Text Solution:

Port Number	TCP/UDP	Protocol
20	TCP	FTP-DATA
21	TCP	FTP-CONTROL
22	TCP	FTP-CONTROL
23	TCP	TELNET
25	TCP	SMTP
67	UPD	DHCP-SERVER
68	UPD	DHCP-CLIENT
80	TCP	HTTP
110	TCP	POP-3
443	TCP	HTTPS

Q7 Text Solution:

The session layer is responsible for dialogue control and synchronisation.

The session layer allows a process to add checkpoints, or synchronisation points, to a stream of data.

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